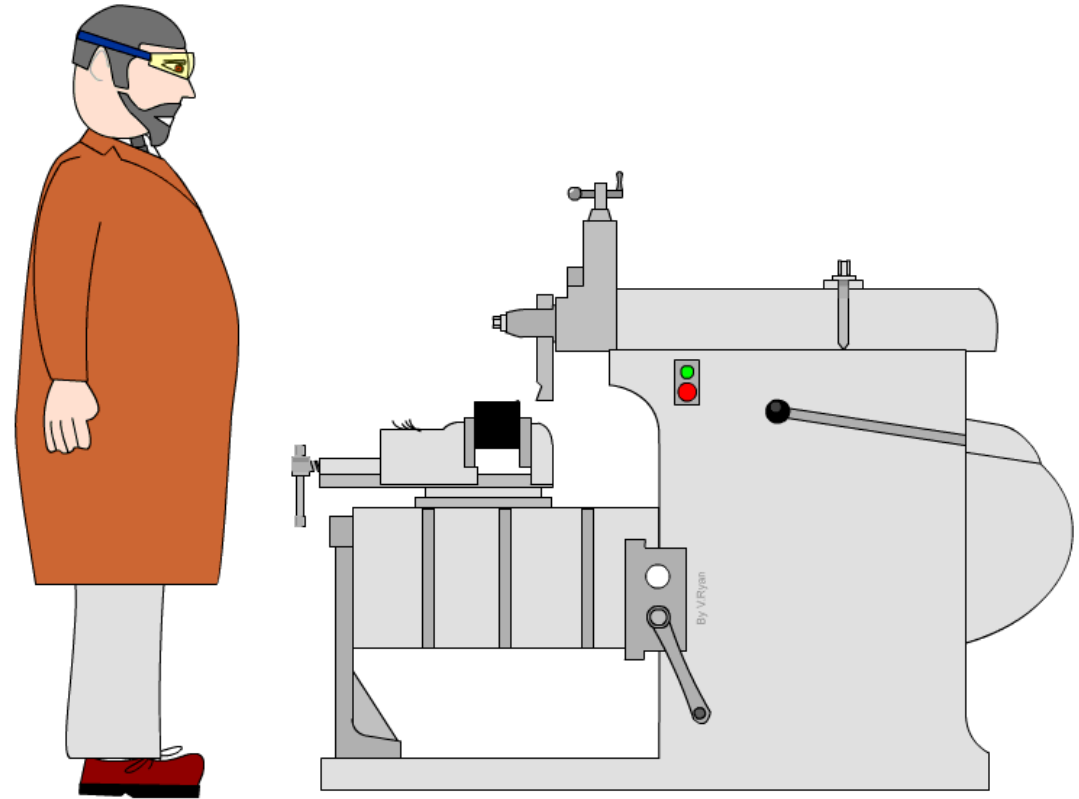


MACHINING PROCESSES

ME1031 – Engineering Skill Development
Department of Mechanical Engineering, Sri Lanka
Institute of Information Technology



GIF courtesy: <https://technologystudent.com/equip1/shape1.htm>

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LEARNING OUTCOMES

- At the end of this section, students should be able to
 - **LO-6: Use of basic hand tools and machine tools in engineering workshop for machining processes and joining metals**

OUTLINE

- Introduction
- Machining operations, tools and equipment
- Pros and cons of machining

INTRODUCTION

INTRODUCTION

- Machine shop is a workshop where machining operations are being performed
 - Machining is a process of removing material from a workpiece by producing chips
 - Material can be metal, plastic, wood or even glass
 - The person who operates the machine → Machinist



Kalpakjian, S., & Schmid, S. R. (2006). Manufacturing Engineering and Technology (6th ed.). Pearson Education South Asia.
Image courtesy: Internet

MACHINING OPERATIONS, TOOLS AND EQUIPMENT

MACHINING OPERATIONS

- Drilling
- Milling
- Turning
- Sawing
- Shaping and Planing
- Grinding



A machining operation

Chapman, W.A.J. (1972). Workshop Technology: Part I. and Part 2. Routledge.

Kalpakjian, S., & Schmid, S. R. (2006). Manufacturing Engineering and Technology (6th ed.). Pearson Education South Asia.

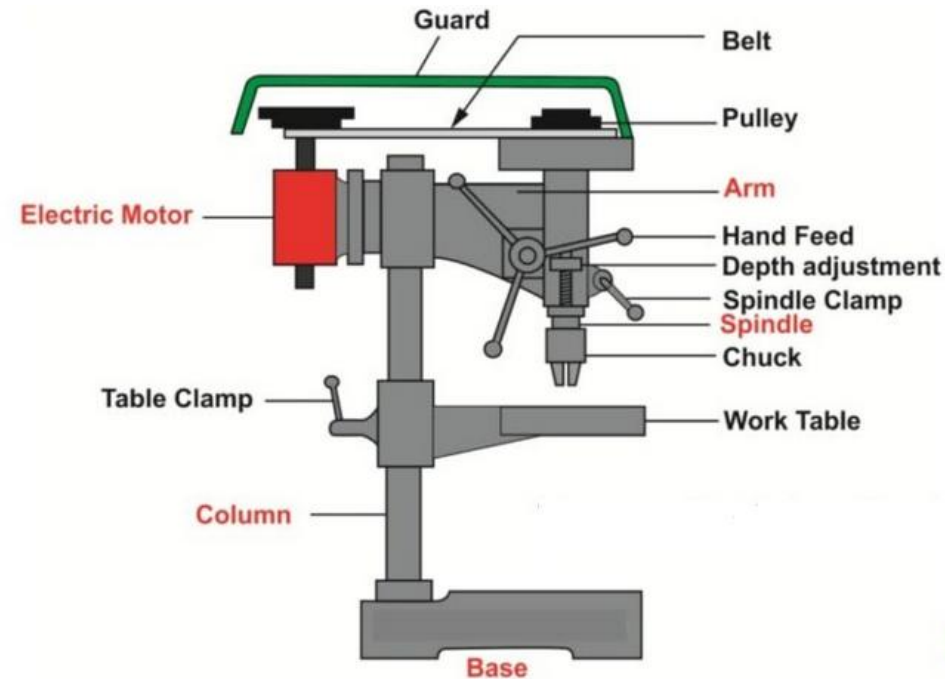
<https://gfycat.com/anxiousagreeableamethystgemclam>

DRILLING

- Used to make holes in a stationary workpiece by a rotating cutter
 - A center punch is used to mark the center of the hole before drilling
- Equipment → Drilling machine
- Cutting tool → Drill bit
- Different drilling operations
 - Drilling
 - Boring
 - Counter boring
 - Counter sinking
 - Tapping



Hand drill



<https://engineeringlearn.com>



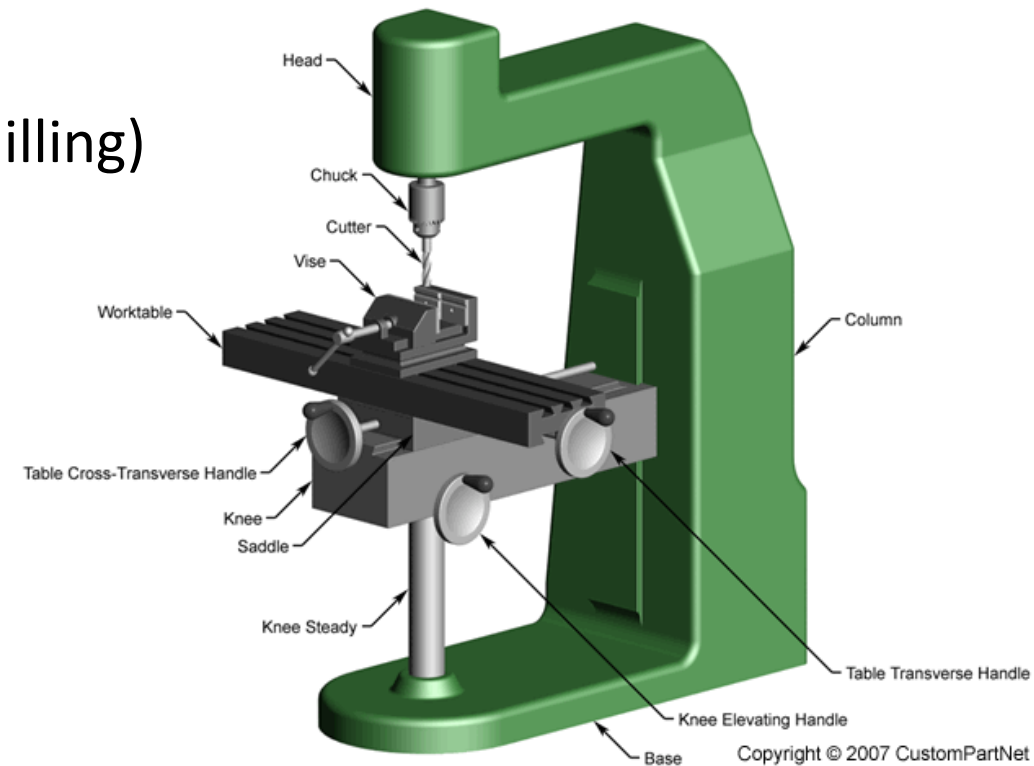
Drill press

Chapman, W.A.J. (1972). Workshop Technology: Part I. and Part 2. Routledge.
<https://en.wikipedia.org/wiki/Drill>
<https://engineeringlearn.com/drilling-machines-types-operation/>
<https://www.youtube.com/watch?v=Gg9OPADuYEo>

MILLING

Used to remove material from a workpiece by advancing the workpiece against a rotating cutter (milling cutter)

- Equipment → Milling machine
- Cutting tool → Different types of milling cutters
- Different milling operations
 - Plain or Slab milling (Two methods: Up / Down milling)
 - End milling
 - Face milling
 - Straddle milling
 - Form milling
 - Angular milling
 - Thread milling
 - Slotting
 - Gear milling

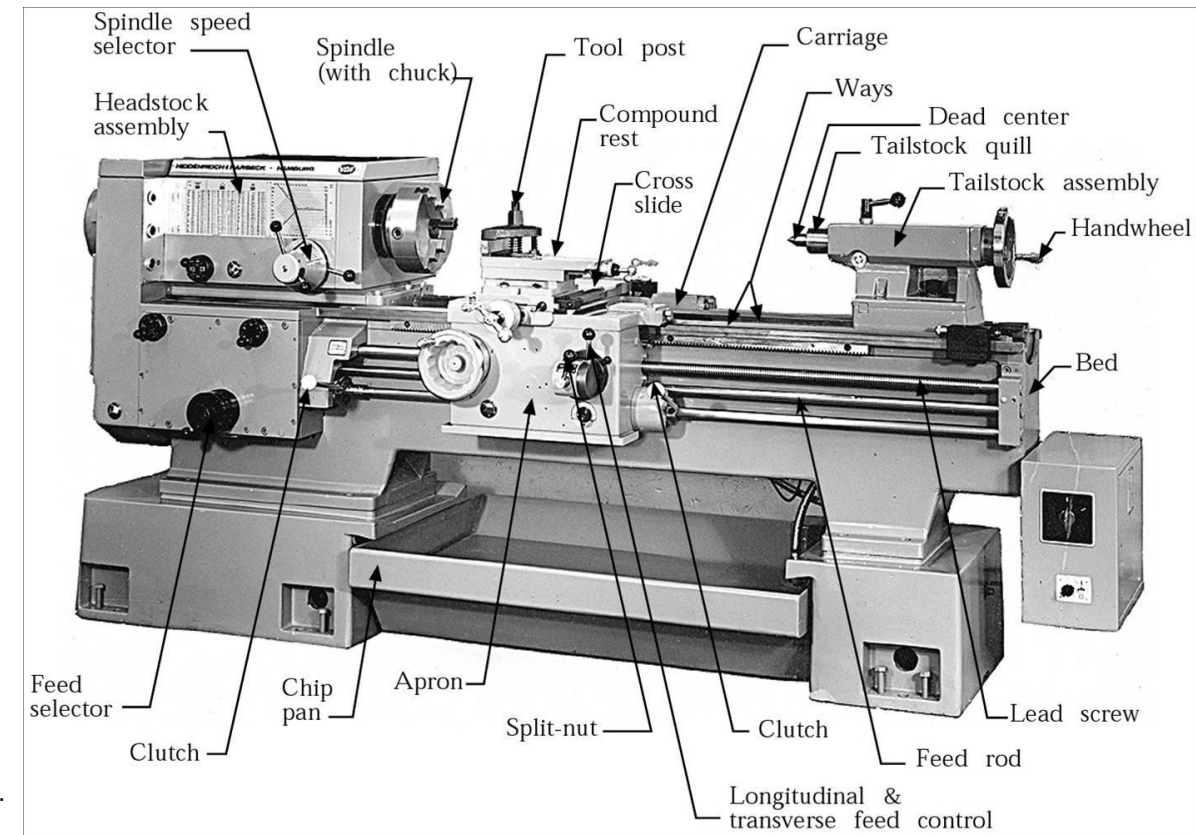


Vertical milling machine

Chapman, W.A.J. (1972). Workshop Technology: Part I. and Part 2. Routledge.
<http://www.custompartnet.com/wu/images/milling/milling-machine.png>
<https://www.youtube.com/watch?v=2jc3HkrHh9s>

TURNING

- Used to remove material from a rotating workpiece by advancing a non-rotating cutter (cutting tool/lathe tool) towards the workpiece
- Equipment → Lathe machine
- Cutting tool → Different types of lathe tools
- Different turning operations
 - Turning
 - Straight
 - Taper
 - Facing
 - Drilling
 - Boring
 - Reaming
 - Parting
 - Grooving
 - Threading
 - Knurling



Lathe machine

Kalpakjian, S., & Schmid, S. R. (2006). Manufacturing Engineering and Technology (6th ed.). Pearson Education South Asia.
<https://www.youtube.com/watch?v=MyAzjhXRWUQ>
<https://www.youtube.com/shorts/KwphFGlo-6I>
<https://www.youtube.com/watch?v=H6Dnmd3IDzA&list=PLY67-4BrEae9Ad91LPRIhcLJM9fO-HJyN>

SAWING

- Used to cut materials by a rotating or a reciprocating toothed edge
- Cutting tool → Blade, chain or a wire
- Different types of saws
 - Hand saw
 - Hacksaw
 - Circular saw
 - Bench saw
 - Jigsaw
 - Band saw
 - Chain saw



Jigsaw



Circular saw



Band saw

Chapman, W.A.J. (1972). Workshop Technology: Part I. and Part 2. Routledge.

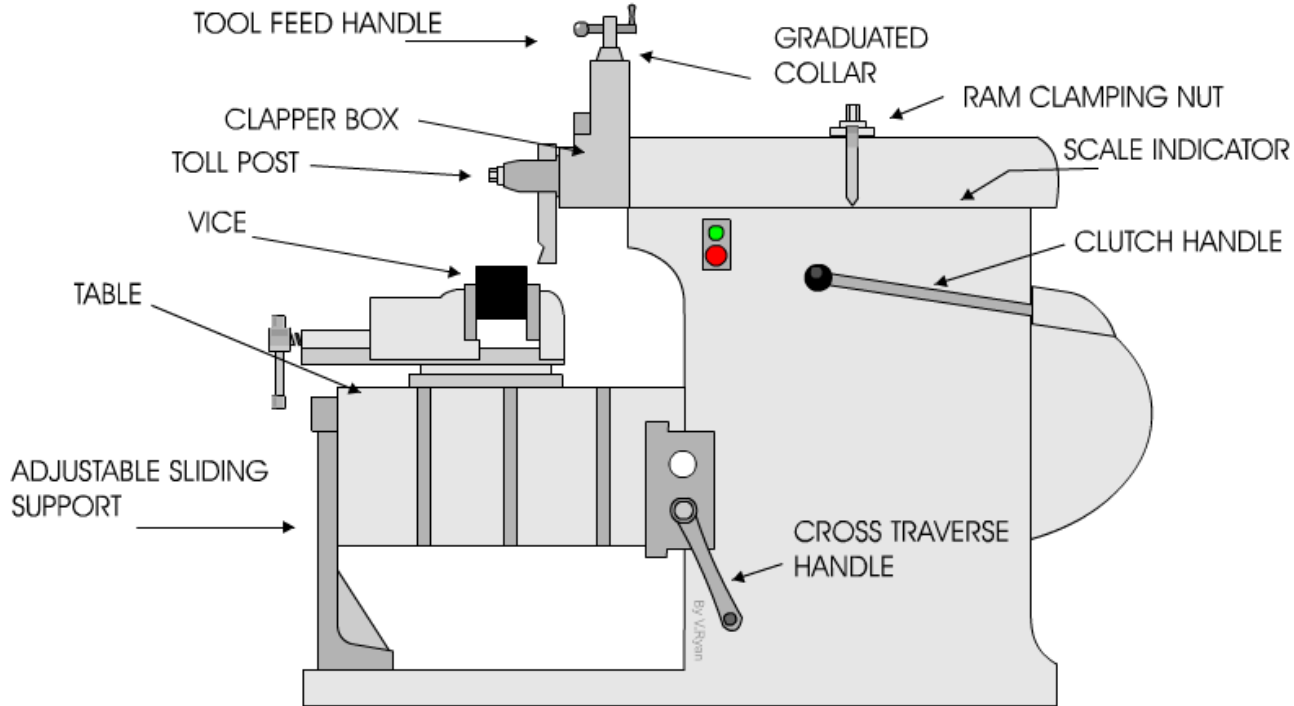
<https://www.boschtools.com/us/en/>

https://www.youtube.com/watch?v=penb1_nuFKM

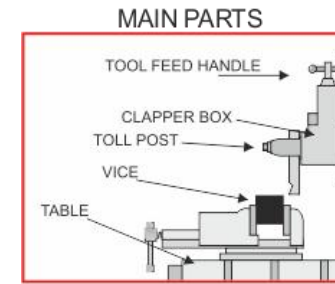
SHAPING AND PLANING

- Used to remove material from the surface of a workpiece to obtain a flat surface
 - Shaping
 - Cutting tool reciprocates against a stationary workpiece → Table where the workpiece is fixed can be moved using the cross-feed (or cross transverse) handle
 - Planing
 - Workpiece reciprocates against a stationary cutting tool → Toolhead where the tool is fixed can be moved using cross rails
 - Usually planing machines are larger than shaping machines
- Cutting tool → Shaping/Planing machine tools

SHAPING MACHINE



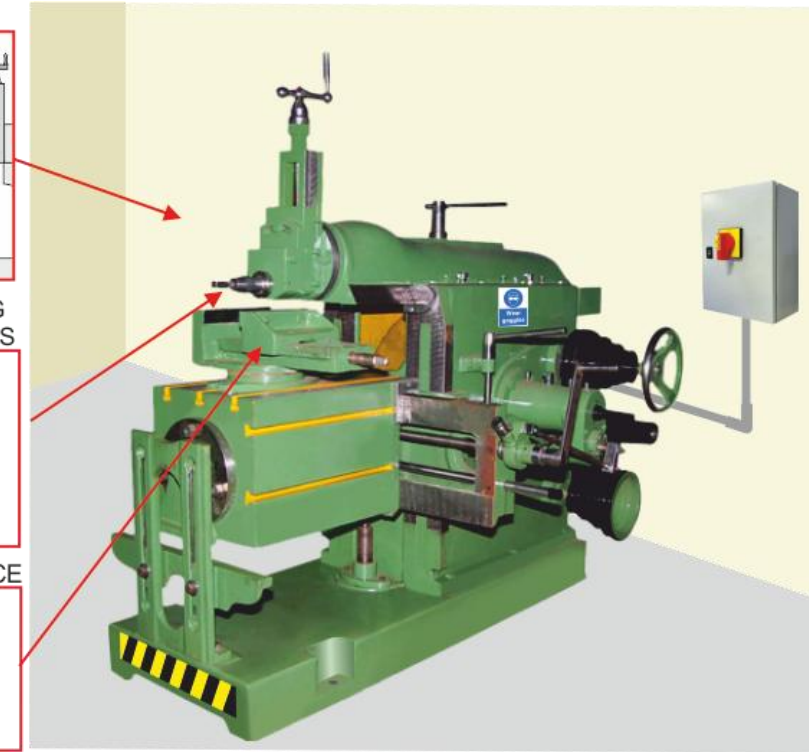
Parts of a shaping machine



SELECTION OF SHAPING MACHINE CUTTING TOOLS

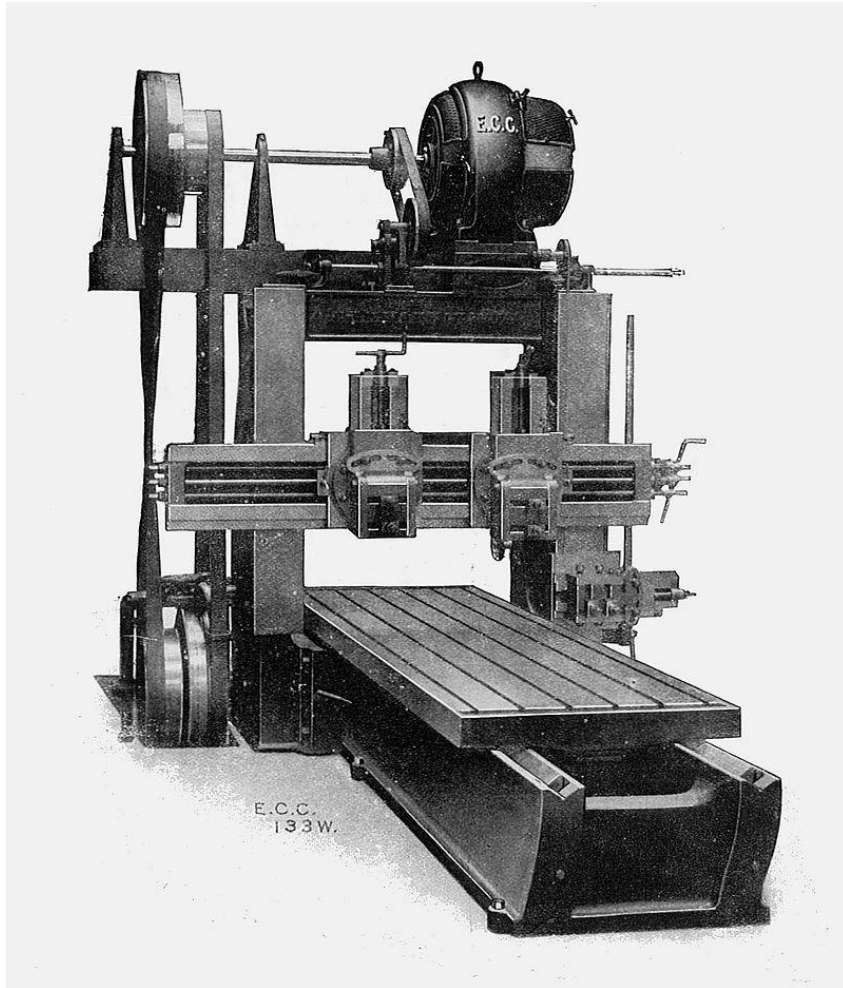


HEAVY DUTY MACHINE VICE

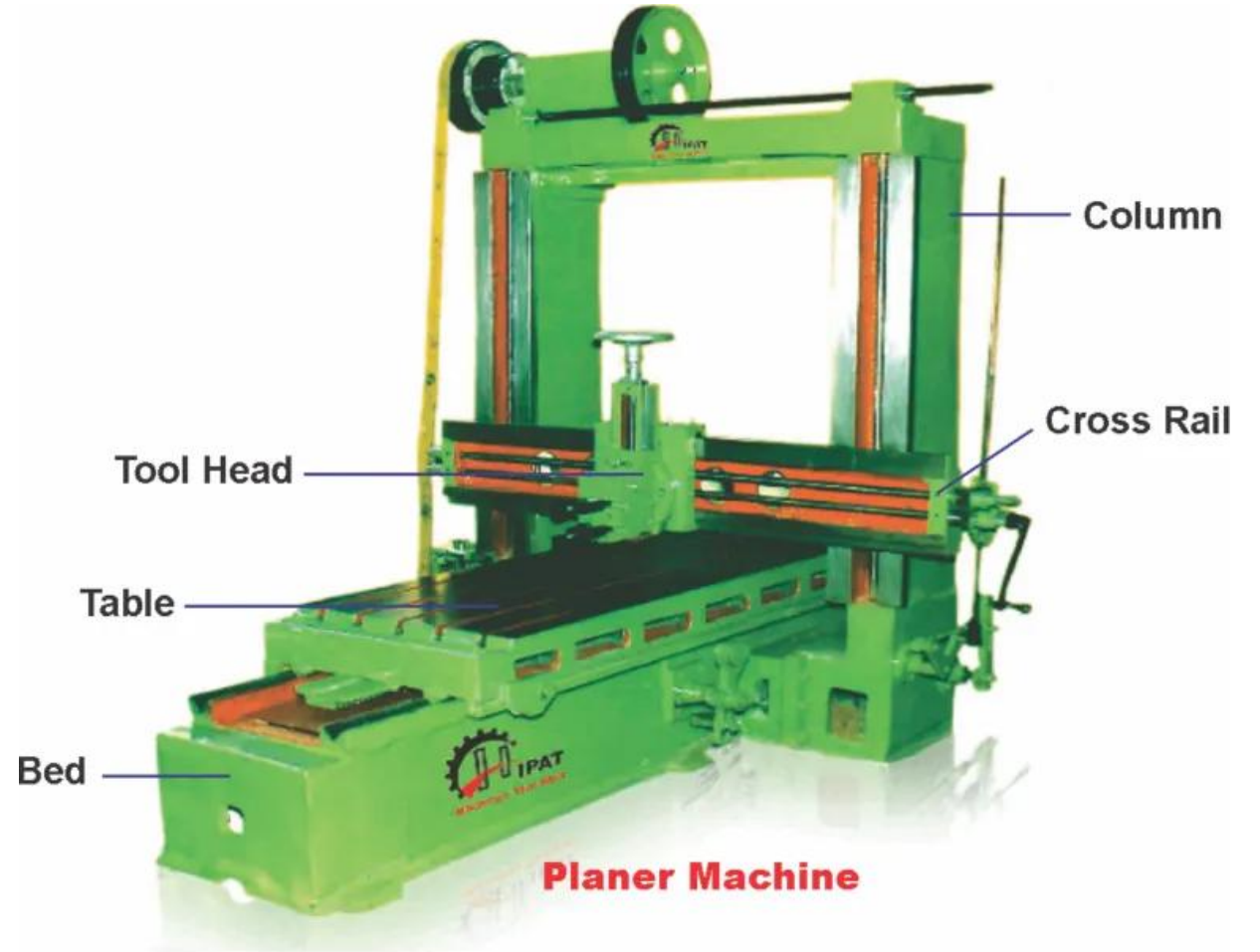


Shaping machine, tools and machine vice

PLANING MACHINE



Planing machine

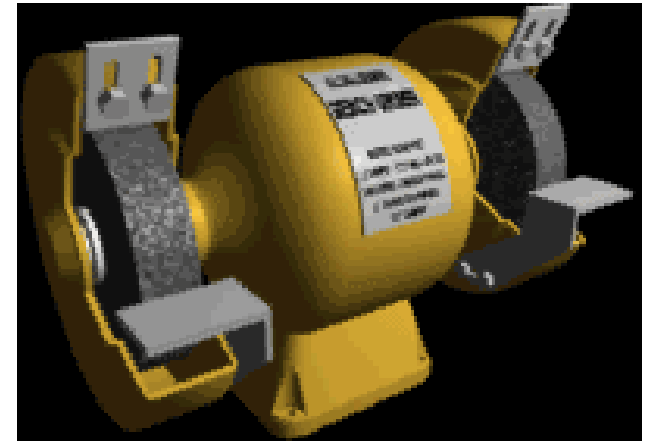


Parts of a planing machine

Chapman, W.A.J. (1972). Workshop Technology: Part I. and Part 2. Routledge.
[https://en.wikipedia.org/wiki/Planer_\(metalworking\)](https://en.wikipedia.org/wiki/Planer_(metalworking))
<https://www.mechanicalbooster.com/2019/10/planer-machine.html>
https://www.youtube.com/watch?v=euovvU_Ojtg&

GRINDING

- Used to remove material from the surface of a workpiece by means of abrasion
- A finishing operation
 - Reduce surface roughness → Obtain a smooth surface
- Tool → Grinding wheel



Bench grinder



Angle grinder



Surface grinder



Cylindrical grinder

<https://www.boschtools.com/us/en/>

https://en.wikipedia.org/wiki/Grinding_machine

<https://www.machinemfg.com/cylindrical-grinder/>

Chapman, W.A.J. (1972). Workshop Technology: Part I. and Part 2. Routledge.

<https://www.youtube.com/shorts/lxtE1eS-07E>

<https://www.youtube.com/shorts/9MvXwbbg90>

PROS AND CONS OF MACHINING

PROS AND CONS OF MACHINING

Pros	Cons
Can produce complex geometric shapes	Material wastage
Can produce dimensionally accurate components	High energy consumption of the machines
Can produce identical components	Heat generation
Good surface characteristics	Undesirable stresses acting on the workpiece
Can be used as a finishing operation	Need skilled labour to operate the machines

We are at the end of this section

- Now you should be able to
 - LO-6: **Use of basic hand tools and machine tools in engineering workshop for machining processes** and joining metals



END